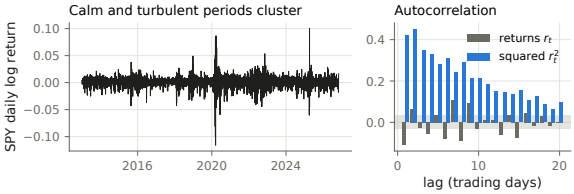
Bayesian GARCH: From Volatility Forecasting to Portfolio Allocation

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Does accounting for Bayesian parameter uncertainty improve volatility forecasts and the performance of investment portfolios based on those forecasts?

Why GARCH, and how do we fit it?



SPY, 2013–26: large moves cluster; squared returns have lag-1 autocorrelation 0.42.

Implication: volatility varies over time and exhibits temporal dependence.

GARCH(1,1): shocks plus persistence

$$r_t = \sqrt{h_t}\, z_t,$$
$$h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}.$$

r_t: daily log return; h_t: conditional variance.

α: response to yesterday's squared-return shock.

β: persistence from yesterday's variance.

ω: baseline; z_t: unit-variance innovation.

Classical — MLE

Maximum likelihood estimation: fit Gaussian GARCH to obtain one parameter vector $\hat{\theta}$.

Bayesian — MCMC

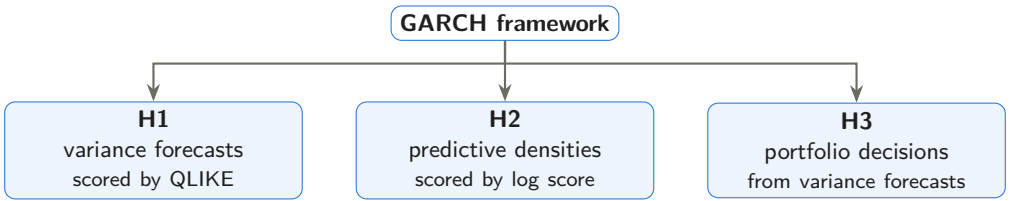
Markov chain Monte Carlo: estimate the Gaussian GARCH posterior and propagate parameter uncertainty into forecasts.

Three evaluations of the GARCH framework

H1 Does Bayesian integration improve one-day-ahead variance forecasts relative to MLE?

H2 Does it improve the full predictive return distribution?

H3 Does posterior uncertainty improve volatility-targeted portfolios?

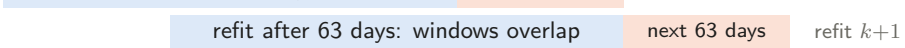


Data and rolling out-of-sample design

SPY, QQQ: US equities **TLT**: long Treasuries **GLD**: gold



Primary: methods frozen before results were inspected. Secondary: previously inspected during development.



Each window ends before its first forecast date; parameters stay fixed for 63 days.

Inside each 63-day block: repeat daily

Past returns $\longrightarrow$ **forecast next day's variance** $\longrightarrow$ observe return $\longrightarrow$ update variance.

Parameters stay fixed between refits; the variance forecast responds to new returns every day.

Every forecast uses only information available before the corresponding return is observed.

H1 How do we evaluate an unobserved variance?

Question: does Bayesian integration improve conditional variance forecasts?

MLE

$$\hat{h}_t^{\mathrm{MLE}} = h_t(\hat{\theta})$$

One estimated parameter vector.

Bayesian integrated forecast

$$\bar{h}_t = \frac{1}{M} \sum_{m=1}^{M} h_t(\theta^{(m)})$$

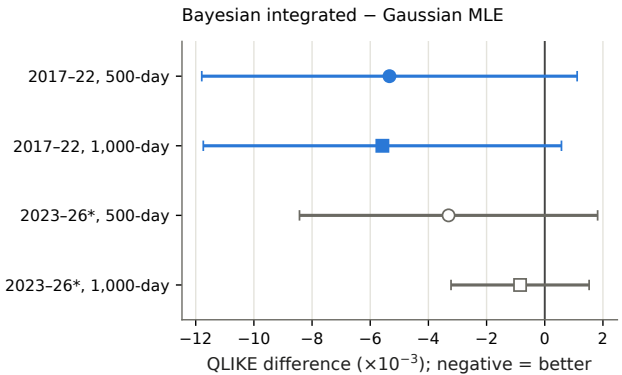
Average forecasts over posterior draws.

Metric: QLIKE (lower is better)

$$\mathrm{QLIKE}_t = \log \hat{h}_t + \frac{r_t^2}{\hat{h}_t}$$

True conditional variance is latent. Under the zero-mean model, squared returns are noisy but unbiased proxies. Expected QLIKE is minimised by the true conditional variance.

H1 Small QLIKE gains, but intervals include zero



Primary: Bayes integrated − MLE

500-day: -0.00534
$p = 0.105$

1,000-day: -0.00558
$p = 0.076$

Interpretation: small numerical improvements, but the confidence intervals cross zero.

95% Newey–West intervals; four-ETF daily averages. * Previously inspected secondary period.

H1: small numerical gains; the intervals cross zero, so no reliable advantage at the 5% level.

H2 — Does Bayesian Integration Improve Predictive Return Densities?

Log predictive score (higher is better):

$$S_t = \log p(r_t^{\text{obs}} \mid D_{t-1})$$

Score the density, fixed beforehand, at the observed return.

MLE

One Gaussian density.

Bayesian mixture

Average Gaussian densities over posterior draws, **then take the log**.

Bayesian improvement over Gaussian GARCH (MLE), 2017–2022

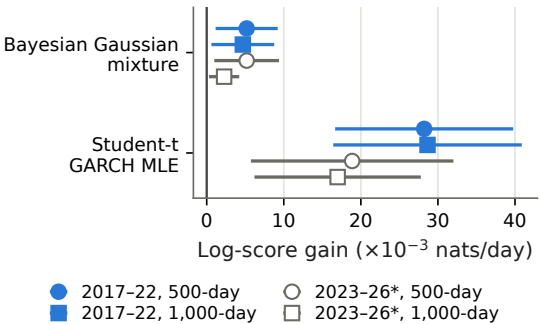
Metric	500-day	1,000-day
Mean log-score gain (nats/day)	$+0.00518$	$+0.00469$
p-value	0.012	0.024

Gains relative to Gaussian GARCH (MLE), not absolute log predictive scores.

Takeaway: Bayesian integration yields small but statistically significant improvements in predictive accuracy.

H2 — Does the Innovation Distribution Matter More?

Compare Bayesian parameter integration with Student-t GARCH (MLE) as an additional benchmark.



Gains over Gaussian GARCH (MLE)
2017–2022, average nats/day

Approach	500-day	1,000-day
Gaussian GARCH (Bayesian)	$+0.00518$	$+0.00469$
Student-t GARCH (MLE)	$+0.02822$	$+0.02865$

95% Newey–West CIs; hollow markers = previously inspected secondary.

Takeaway: Student-t GARCH delivers approximately 5–6× the log-score improvement of Bayesian parameter integration under Gaussian GARCH.

H3 From variance forecasts to ETF/cash portfolios

Volatility targeting: 10% annual target, no leverage.

$$w_d = \min\left(1, \frac{0.10}{\sqrt{252\, v_d}}\right)$$

ETF share w_d; cash share $1 - w_d$.

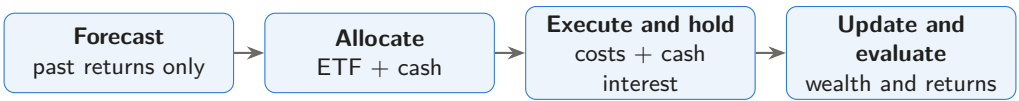
Higher predicted volatility ⇒ less ETF exposure.

Two forecasts for the initial comparison

Standard MLE: $v_d = h_d(\hat{\theta})$.

Bayesian upper-90: $v_d = q_{0.9,d}$.

Bayesian upper-90 uses the posterior 90th percentile of conditional variance forecasts rather than the MLE plug-in forecast, generally producing lower ETF exposure.



Repeat daily. Execute one close after the signal; 5 bp × turnover; cash earns DFF.

H3 Initial result: safer portfolios, but less exposure

2017–22

Strategy	Exposure	Return	Volatility	Sharpe	Max DD
Standard MLE	73.1%	6.25%	10.37%	0.518	−19.55%
Bayesian upper-90	69.6%	6.06%	9.81%	0.520	−18.74%

Means of 8 single-ETF configurations (4 ETFs × 2 windows); return/volatility annualised.

Volatility falls and Sharpe is nearly unchanged. Paired 95% intervals (20-day block bootstrap) exclude zero for the drawdown difference in 8/8 configurations, for the Sharpe difference in 0/8.

Main observation

Bayesian upper-90 reduces maximum drawdown, but also holds about **3.6 pp less risky exposure**.

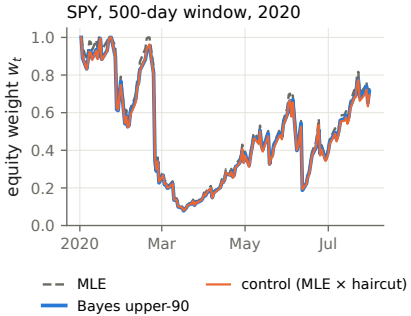
Does Bayesian uncertainty improve allocation timing, or does the lower drawdown simply reflect holding less risk?

H3 Control for exposure before claiming better timing

Exposure-matched MLE: keep the MLE variance forecasts; adjust only the effective volatility target:

$$w_t^C = \min\left(1, \frac{k_t\, 0.10}{\sqrt{252\,\hat{h}_t^{\mathrm{MLE}}}}\right)$$

k_t: trailing mean of past forecast ratios $\sqrt{\hat{h}_s^{\mathrm{MLE}}/q_{0.9,s}}$ (posterior upper-90), so average ETF exposure approximately matches upper-90's; daily risk need not be identical.



The exposure-matched MLE strategy closely tracks Bayesian upper-90 allocations, suggesting that a simpler exposure adjustment can reproduce much of its conservative behavior.

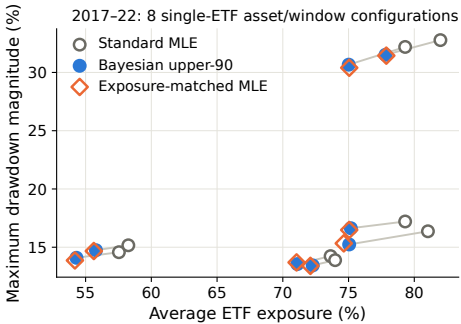
H3 After exposure matching, outcomes are very similar

Strategy	Exposure	Return	Volatility	Sharpe	Max DD
Exposure-matched MLE	69.4%	6.01%	9.85%	0.516	−18.66%
Bayesian upper-90	69.6%	6.06%	9.81%	0.520	−18.74%

Volatility, drawdown and Sharpe ratio are very similar.

Paired 95% intervals for the Sharpe and drawdown differences include zero.

2017–22; means of 8 single-ETF asset/window configurations.



After approximately matching average risky exposure, portfolio risk and performance are very similar. No statistically reliable incremental benefit from Bayesian upper-90 is established.

Conclusions: forecasting gains and investment evidence

H1 **Variance:** small QLIKE gains; no reliable advantage at 5%.

H2 **Density:** modest Bayesian gains; Student-t MLE improves substantially more.

H3 **Portfolios:** lower exposure and drawdowns, but no reliable incremental benefit versus exposure-matched MLE.

Finding 1: innovation-distribution specification is highly consequential for density accuracy.

Finding 2: apparent portfolio improvements need controls for differences in risky exposure.

Better statistical forecasts do not automatically translate into better investment performance.